Gradient-Sensitive Functional Descriptors for Interferometric Roundness Metrology: Complementing Amplitude-Averaging ISO Parameters

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Abstract

Amplitude-averaging surface parameters such as R_a may show limited response to localised defects — a controlled sensitivity experiment indicates that a single 100 nm synthetic spike injected into a spherical-artefact profile produces no detectable change in R_a at the reconstruction resolution, and the ISO 4287 shape parameters R_{sk} and R_{ku} show similarly limited response to such symmetric localised features. This paper evaluates the same reconstructed profile data within a functional-analytic norm framework — L^p, Sobolev $W^{1,2}$, and bounded-variation (BV) norms, standard tools of functional analysis here applied within a metrological processing pipeline — finding higher sensitivity to such defects (BV density: +18%; $W^{1,2}$ gradient seminorm: +622%) without additional hardware. A complete proof-of-concept pipeline from raw Michelson-interferometer fringe images to nanometre profile descriptors is presented and validated on two JENOPTIK standards (roundness standard FN 111, $\varnothing = 29.9588$ mm, 24,000 frames; magnification standard FN 101 with Flick flat, 100,000 frames), each measured in five 400° sweeps. External validation against 104 tactile reference measurements (Hommel Roundscan 535) yields $E_n = 0.61$ for the FN 111 and 0.3% agreement for the FN 101 (optical–tactile polar comparison: Supplementary Fig. S12), suggesting metrological consistency of the optical pipeline within the stated within-run type-A uncertainties. The functional-norm quantities are presented as complementary to the ISO 4287 baseline: R_{sk} responds to asymmetry, BV density responds to localised gradients, and their combination may provide descriptive sensitivity beyond that of any single classical parameter in the tested scenarios. The central finding is that BV density detects a certified geometric feature (the Flick flat on the FN 101 standard) that remains below the detection threshold of amplitude-averaging parameters in a sliding-window analysis matched to the feature's certified angular extent, illustrating — in the tested proof-of-concept scenarios — that gradient-based quantities can provide complementary descriptive sensitivity to localised geometric features without additional hardware; the L^4/L^1 peakiness ratio (1.39 ± 0.08 for the FN 111, 1.53 ± 0.03 for the FN 101; difference not statistically distinguishable at 95% confidence given overlapping expanded uncertainties) is reported as an illustrative distribution-shape indicator reflecting the amplification of dominant profile excursions relative to R_a.

Keywords: interferometry, roundness, surface profile, Banach-space norms, bounded variation, Sobolev seminorm, uncertainty

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1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement [7–11], while interference remains central for ultra-precise surface-form measurement [12, 13] and phase-shifting interferometry is widely used in topography evaluation [14–16].

Several interferometric instruments have been proposed for high-accuracy surface and roundness measurement. Polack et al. described a Michelson-based system for surface-shape determination built around a high-resolution stitching interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented a roundness measuring system for microspheres based on two small Michelson interferometers [19]. Kuhnel et al. proposed an interferometric system for roundness and cylindricity measurement of ring gauges with 0.1 nm resolution over a large measurement range [20]. Commercial instruments can be highly accurate, but they are often costly and may not expose the full acquisition chain needed for experimental method development. The present work therefore uses a self-built, non-contact Michelson-type roundness measurement system assembled from off-the-shelf components. The optical setup, apparatus description and radial image-processing route for phase extraction were published in a previous study [21], where only roughness parameters were reported for the ceramic standard FN 111. The present work extends that study in three directions: (i) roundness ($RONt$, $RONq$) is evaluated for the first time with this self-built optical setup, (ii) a second artefact — the JENOPTIK magnification standard FN 101 with a Flick flat — is measured and validated against its DAkkS-DKD certificate, and (iii) a controlled sensitivity-comparison methodology is introduced: synthetic defects of known geometry are injected into measured profiles, and the response of classical ISO parameters is compared against that of L^p, Sobolev $W^{1,2}$, and bounded-variation (BV) norms — standard tools of functional analysis here applied to interferometric profile metrology — all computed from the same nanometre residual. The comparison framework, rather than any individual norm, constitutes the methodological contribution.

The computational problem is that modern interferometric surface metrology records long image sequences before a numerical profile exists. In the present data, however, each PNG file is already a separate interferogram in which the phase is encoded spatially by the bright and dark fringes inside the image. The files in the FN 111 and FN 101 folders are therefore not treated as a temporal phase-shifting stack of one fixed scene. Instead, each image $I_i(x, y)$ is processed independently to obtain one wrapped excess-fraction value ϵ_i. The ordered sequence of these values is then unwrapped over image order and converted to a nanometre profile. The full measurement archive contains 24,000 grayscale interferograms of the FN 111, with image size 648×488 px, and 100,000 grayscale interferograms of the JENOPTIK magnification standard FN 101, with image size 640×480 px.

The literature relevant to this comparison is mature but only partly connected. Interferometric testing has developed approaches for stitching, registration, motion-error compensation, and systematic-error control [22–26]. Repeatability is commonly evaluated through RMS/PV quantities or pixel-wise standard-deviation matrices [27, 28]. Uncertainty-oriented surface metrology embeds surface-texture parameters in GUM-style propagation [29]. Functional data analysis provides a statistical language for function-valued observations [30, 31], and scale-dependent roughness studies show that characterisation can depend on bandwidth and filtering scale [32–34].

The research gap is therefore specific. Existing interferometric literature provides robust instrumentation and phase reconstruction; standards-oriented work provides scalar profile and form parameters with uncertainty evaluation; and functional data analysis provides a mathematical language for function-valued observations. What has received less attention is a controlled, within-profile comparison of the sensitivity of classical ISO amplitude parameters against variation-based functional norms, computed from the same nanometre residual and validated against an external tactile reference. The present work addresses this by: (i) injecting synthetic defects of known geometry into measured profiles to quantify descriptor response under controlled conditions; (ii) applying the full pipeline to two DAkkS-DKD-certified artefacts at large scale (24,000 and 100,000 interferograms) with a self-built radial excess-fraction interferometer; and (iii) benchmarking the optical reconstruction against 104 tactile reference measurements. Areal parameters are not reported because the single-image radial extraction does not produce a calibrated areal height map. The aim is to assess whether L^p, bounded-variation, and Sobolev seminorms, evaluated alongside ISO 4287 parameters on the same reconstructed residual profile, offer complementary descriptive sensitivity to controlled profile perturbations, and to evaluate the within-run repeatability of all quantities across five recorded sweeps.

2. Experimental Details

2.1. Optical Setup

The optical layout of the measurement system is shown in Fig. 1. The system is a two-channel modified Michelson interferometer with channels shifted by approximately 90° in phase. When an array detector is used instead of a point detector, the second channel is not strictly required [35]. In this system it was retained because it is useful for object centring and can serve as a control channel.

The optical setup and the full apparatus-level description were reported in a previous study on radial image processing for phase extraction in rough-surface interferometry [21]. They are repeated here in condensed form to make the present functional-descriptor analysis self-contained and to document the acquisition conditions behind the FN 111 and FN 101 interferogram archives.

The system was first tested with the JENOPTIK roundness standard FN 111 (photographs: Supplementary Fig. S1) (Al_2O_3, $\varnothing = 29.9588$ mm) [36]. A single-mode 4.5 mW diode laser with wavelength 532 nm was used as the light source. A current-stabilised supply powered the laser to obtain a stable wavelength [37]. The laser beam passes through a polariser (P in Fig. 1) to set the polarisation of the light at 45° to the setup plane, minimising the difference in reflectivity between the two channels.

Key components: achromatic lenses $L_{1,2,4,5}$, non-polarising beamsplitter BS, $\lambda/8$ liquid phase retarder, $\lambda/10$ reference mirror M, DIN-4 microscope objective L_3 (NA 0.10), polarising beamsplitter PBS, and two Gigabit Ethernet CCD cameras (640×480 px, 120 fps).

The interferometer was built in a cage system for thermal, acoustic and mechanical stability. A plexiglass box shields the measurement object, and the

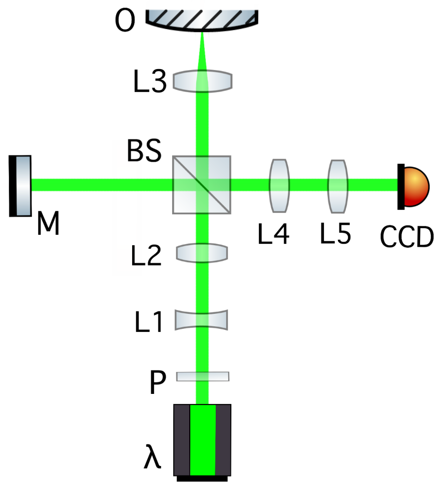

Figure 1: Optical layout of the two-channel modified Michelson interferometer: λ – diode laser (532 nm); P – linear polariser; L – lenses; L_3 – microscope objective; M – reference mirror ($\lambda/10$ flatness); BS – non-polarising beamsplitter; O – object on rotation stage; CCD – camera.

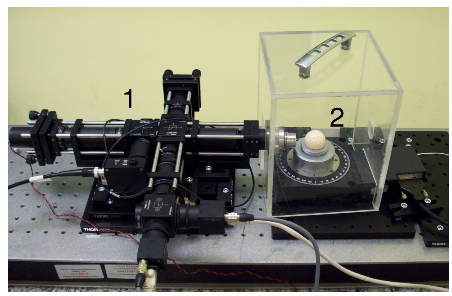

Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

structure is mounted on an area translation stage. A high-precision rotation stage (encoder accuracy 2.0 mdeg, typical wobble ± 12 μrad, typical eccentricity ± 0.40 μm; full specifications: Supplementary Table S1) rotates the object. Both interferometer and stage are mounted on a honeycomb breadboard with vibration isolation.

Measurements were performed at night (23:00–05:00) to minimise building-activity disturbances. The setup was thermally and acoustically shielded (plexiglass box, wool blanket) and placed on a granite plate on layered wooden-polystyrene vibration isolation (photographs: Supplementary Fig. S2). Five semiconductor temperature sensors (± 0.2°C) monitored stability; the temperature near the reference and object arms remained stable to ± 0.2°C over six-hour acquisitions.

2.2. Control System and Acquisition Workflow

Cameras were controlled by FLIR SDK-based C programs and Bash scripts [38]. Capture time was recorded per frame for angular-position verification. The rotation stage was controlled via USB ASCII commands [39]; a 25 s settling delay ensured constant velocity before acquisition began. Object centring was performed manually using the two-camera phase-shifted configuration at 45° rotation increments. Fringe evaluation used modified open-source R code [35, 40]. Table 1 lists the acquisition parameters for both archives; five 400° sweeps were recorded per artefact, with the 40° overlap retained for profile-end comparison.

Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

Parameter	FN 111	FN 101
Rotation speed, ω	0.50 °/s	0.08 °/s
CCD frame rate	6 fps	4 fps
Recorded sweeps, n	5	5
Total frames, N	24,000	100,000
Angular step, $\Delta\alpha = \omega/\text{fps}$	0.083°/frame	0.020°/frame
Frames per degree	12	50
Frames per useful 360°	4320	18,000
Frames per recorded 400° sweep	4800	20,000
Overlap per sweep	40° / 480 frames	40° / 2000 frames
Total acquisition time	4000 s	25,000 s

The object was measured in one continuous five-sweep rotation without repositioning. Each recorded sweep spans 400°; successive useful 360° segments are shifted by 40°, giving a natural between-sweep angular offset. The 40° overlap region provides a model-free repeatability check: RMS height difference between consecutive sweeps is 12 nm (FN 111) and 0.18 μm (FN 101), consistent with Table 2.

3. Image Processing and Functional-Descriptor Method

3.1. Processing Pipeline

The pipeline follows the radial excess-fraction method of a previous study [21] with added sinusoidal refinement. Each interferogram $I_i(x, y)$ is decoded independently to a wrapped excess fraction ϵ_i; the sequence is unwrapped chronologically, converted to nanometres ($z_i = \nu_i\lambda/2$), and reduced by the object-specific LSCI reference model. Sweep boundaries are retained for repeatability analysis.

3.2. Step-by-Step Radial Phase Decoding

For each interferogram $I_i(x, y)$, a radial intensity profile about the fringe centre is extracted and B-spline-smoothed; bright-ring maxima are detected in D^2 space, where circular-fringe theory gives $D_{ij}^2 = c_i(j + \epsilon_i - 1)$ for the jth ring of diameter D_{ij}. A linear fit $D_{ij}^2 = \alpha_i j + \beta_i$ yields the wrapped excess fraction $\epsilon_i = (1 + \beta_i/\alpha_i) \bmod 1$ (2). Where the radial fringe signal permits ($R^2 \geq 0.65$, phase within 0.20 fringe of the ring-diameter estimate), a sinusoidal refinement is applied; otherwise the ring-diameter estimate is retained. The ordered series ϵ_i is unwrapped chronologically to a continuous fringe-order coordinate ν_i, and the nanometre profile follows from $z_i = \nu_i\lambda/2$ with $\lambda = 532$ nm (one fringe order $\equiv$ 266 nm) (3).

$$D_{ij}^2 = c_i(j + \epsilon_i - 1), \tag{1}$$

$$\epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i}\right) \bmod 1. \tag{2}$$

$$z_i = \nu_i \frac{\lambda}{2}, \qquad \lambda = 532 \text{ nm}. \tag{3}$$

3.3. Topography Profile, Reference Removal and Eccentricity Reduction

After phase-to-height conversion, the roundness profile is extracted following ISO 12181-1 [41] (which supersedes the withdrawn ISO 4291). For each 360° segment, a least-squares reference circle (LSCI) is fitted to the angular height profile $z_i = h(\theta_i)$ in Cartesian coordinates:

$$(x_i, y_i) = \big((R_0 + z_i)\cos\theta_i,\ (R_0 + z_i)\sin\theta_i\big), \tag{4}$$

where $R_0 = 14.9794$ mm is the nominal radius of the JENOPTIK FN 111. The LSCI centre (x_c, y_c) minimises $\sum_i[\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$, yielding the centred radial residuals ρ_i. The LSCI fit is numerically equivalent to removing the first harmonic of the angular profile, but it provides the formal ISO 4291 reference circle required for roundness metrology. After LSCI centring, a phase-correct Gaussian low-pass profile filter with 50% transmission at 15 undulations per revolution (UPR) is applied to the residual profile, consistent with the filter specification used for the tactile reference measurements and with the Gaussian profile filter defined in ISO 16610-21. All classical and gradient-based descriptors are computed from this LSCI-centred, Gaussian-filtered residual $\rho_i^{(\mathrm{filt})}$.

For the JENOPTIK magnification standard FN 101, the same processing chain is applied: a linear trend is removed from the unwrapped phase sequence before height conversion (Sec. 5.1), the LSCI reference circle is fitted to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is applied. Although both artefacts are processed through identical filtering steps, the 15 UPR cutoff corresponds to different spatial wavelengths on the circumference: $2\pi \times 14.9794$ mm$/15 = 6.27$ mm for the FN 111 ($\varnothing = 29.9588$ mm) and $2\pi \times 15$ mm$/15 = 6.28$ mm for the FN 101 ($\varnothing \approx 30$ mm) — effectively identical. The angular sampling differs substantially (0.083° vs. 0.020°, Table 1), but the Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so the filtered profiles are directly comparable.

3.4. Classical Profile and Roundness Metrics

All classical quantities are evaluated on nanometre residuals, not on raw intensities, following the profile method of ISO 4287 [42]. Areal parameters (ISO 25178 [43]) are not reported because the single-image radial extraction does not produce a calibrated areal height map. For a residual profile or vector of residual pixels p, the reported profile-style roughness descriptors are

$$R_a = \frac{1}{M}\sum_{j=1}^{M}|p_j - \bar{p}|, \qquad R_q = \left[\frac{1}{M}\sum_{j=1}^{M}(p_j - \bar{p})^2\right]^{1/2}, \tag{5}$$

and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \tag{6}$$

The profile skewness R_{sk} and kurtosis R_{ku} — ISO 4287 shape parameters that characterise asymmetry and peakedness of the height distribution — are also reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^{M} (p_j - \bar{p})^3, \qquad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^{M} (p_j - \bar{p})^4. \tag{7}$$

$R_{sk} = 0$ for a symmetric distribution; $R_{sk} < 0$ indicates a predominance of valleys over peaks. $R_{ku} = 3$ for a normal distribution; $R_{ku} > 3$ indicates a spikier (leptokurtic) profile. These are the only ISO 4287 parameters that capture distribution shape beyond amplitude averages, and their inclusion allows a direct test of whether gradient-based descriptors provide information beyond even the best ISO shape parameters. For the eccentricity-corrected ceramic angular profile $q_C(\theta)$,

$$RONt = \max_\theta q_C(\theta) - \min_\theta q_C(\theta), \qquad RONq = \left[\frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta)\, d\theta \right]^{1/2}. \tag{8}$$

These are profile/form quantities in nanometres. B-spline pre-smoothing (20 degrees of freedom, $\approx 18^\circ$ period) stabilises the LSCI fit; the subsequent Gaussian low-pass filter (50% at 15 UPR, ISO 16610-21) defines the metrological bandwidth. All descriptors are computed from the same Gaussian-filtered residual.

3.5. Functional Gradient-Based Descriptors

The concept of a complete normed vector space originates from the foundational work of Banach [44]. For discrete profile data of finite length N, the vector space $\mathbb{R}^N$ equipped with any norm is a Banach space; the present work does not invoke infinite-dimensional completeness but draws on the taxonomy of L^p, Sobolev, and bounded-variation norms that Banach's framework systematised. For a centred residual profile $p(s)$ and $1 \le p < \infty$,

$$\|p\|_p = \left(\frac{1}{L} \int_0^L |p(s)|^p \, ds \right)^{1/p}, \qquad \|p\|_\infty = \max_{s \in [0,L]} |p(s)|. \tag{9}$$

The reported functional descriptors include L^1, L^2, L^∞, the one-dimensional Sobolev $W^{1,2}$ seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left(\int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \qquad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \tag{10}$$

In discrete form, for a uniformly sampled profile p_j $(j = 1, \ldots, N)$ with angular spacing $\Delta\alpha$,

$$|p|_{W^{1,2}} \approx \left(\Delta\alpha \sum_{j=1}^{N-1} \left(\frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \qquad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \tag{11}$$

Both $W^{1,2}$ and BV density carry sampling-interval dependence and require conversion to angular units ($\mathrm{nm\,deg}^{-1/2}$ and $\mathrm{nm\,deg}^{-1}$, respectively) for cross-instrument comparison; "$\mathrm{nm\,sample}^{-1}$" is retained for the fixed-acquisition setup.

The curvature RMS (root-mean-square of the discrete second derivative; reported in Supplementary Table S2) carries a $(\Delta\alpha)^{-2}$ sampling dependence and should not be compared across instruments with different angular resolution.

3.6. Repeatability Uncertainty

Each dataset contains five 400° sweeps recorded in a single continuous rotation without stopping or repositioning. The five useful 360° segments are extracted from this continuous sequence; because each recorded sweep spans 400°, successive useful revolutions are shifted by 40° on the object. These segments are *not* strictly independent — they share the same continuous-run conditions (laser warm-up state, stage-bearing temperature, ambient drift). The standard deviation across sweeps therefore reflects within-run repeatability rather than between-run reproducibility. For a descriptor X evaluated on each sweep, the mean value is $\bar{X}$ and the within-run standard uncertainty is estimated as

$$u_{\mathrm{within}}(X) = \frac{s_X}{\sqrt{n}}, \qquad n = 5, \tag{12}$$

where s_X is the sample standard deviation across the five sweeps. The reported expanded uncertainty is

$$U_{95}(X) = t_{0.975,n-1}\, u_{\mathrm{within}}(X). \tag{13}$$

With $n = 5$, $t_{0.975,4} = 2.776$; the χ^2_4-based 95% CI for the standard deviation spans $[0.6s_X, 2.9s_X]$, so U_{95} is indicative rather than definitive. Because the five segments share continuous-run conditions (laser state, stage temperature, ambient drift), the $1/\sqrt{n}$ factor in u_{within} may underestimate the true uncertainty of the mean; all reported uncertainties are *lower bounds*. A full GUM-compliant budget (type-B terms: laser wavelength stability, refractive-index variations, Abbe error [45], camera nonlinearity, form-removal uncertainty) is beyond the present scope. With $n = 5$ sweeps and the observed within-run variability, the minimum detectable relative difference between two descriptors (80% power, $\alpha = 0.05$, two-sided paired t-test) is approximately 30% for the FN 111 and 5% for the FN 101, consistent with the exploratory character of the descriptor comparison. A per-sweep trend of FN 111 *RONt* values gives a slope of 8.4 ± 4.1 nm sweep^{-1} ($p = 0.12$), consistent with zero drift, though the sweep-to-sweep scatter cautions that U_{95} may underestimate long-term reproducibility.

3.7. Computational Implementation

The reconstruction workflow (`scripts/reconstruct_radial_profile_sequence.py`) decodes ϵ_i from each interferogram, unwraps the sequence, converts to nanometres, applies LSCI fitting and Gaussian low-pass filtering (50% at 15 UPR, ISO 16610-21), and exports CSV/JSON metrics with sweep-repeatability uncertainty. Vector PDF figures document each processing stage. The pipeline is invoked as `PYTHONPATH=src.venv/bin/pythonscripts/reconstruct_radial_profile_sequence.py` with dataset-specific parameters (full command in the data repository). Per-image excess-fraction decoding and unwrapping require approximately 0.5 ms per frame (measured on an Intel Core i7-9700K, single-threaded Python 3.9), so the full 124,000-frame archive processes in under two minutes on consumer hardware.

4. Results

The results follow the updated radial sequence route only. Each raw image is first decoded to a wrapped excess fraction from its radial fringe pattern. The ordered excess-fraction series is then unwrapped in chronological order, converted to nanometres, and corrected for eccentricity. Finally, classical profile/roundness descriptors and functional gradient-based descriptors are computed from the same nanometre residual profiles.

4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

The FN 111 folder contained 24,000 interferograms and the steel folder contained 100,000 interferograms. The reported run used full-resolution images, the central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of the radial fringe-signal range. All five recorded sweeps were processed.

Figs. S3–S6 (single-image EFM and sequence processing steps) are provided in the Supplementary Materials.

4.2. Roughness, Roundness and Functional Descriptors

Table 2 gives the numerical comparison after LSCI reference circle and Gaussian low-pass filtering (50% at 15 UPR). Both artefacts are processed through the same filtering chain. Each descriptor is reported as the full-sequence value together with the 95% expanded type-A repeatability uncertainty estimated from the five sweeps. Per-sweep values for all descriptors are provided in Supplementary Table S3.

The numerical identity $RONt = R_z$ for both artefacts (Table 2) reflects the fact that after LSCI centring and Gaussian low-pass filtering (50% at 15 UPR), the residual profile is dominated by a single lobe (harmonic decomposition: Supplementary Fig. S19); the peak-to-valley roundness deviation and the profile roughness peak-to-valley therefore coincide. This identity is specific to the present artefact–filter combination and should not be expected to hold for artefacts with multi-lobed form deviations or for different filter cutoffs.

4.3. Classical–Functional-Descriptor Comparison

The numerical identity $R_a = L^1$ and $R_q = L^2$ (Table 2) confirms that both descriptor families are evaluated on the same nanometre residual. For the

Table 2: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ($n = 5$, $t_{0.975,4} = 2.776$). Units: nm for all descriptors except BV density and $W^{1,2}$ gradient (nm sample^{-1}). Certified $RONt$: FN 111 244 ± 52 nm (Hommel Roundscan 535, $n = 104$); FN 101 41.94 μm (form tester, $n = 12$). The FN 101 certificate does not report an expanded uncertainty; therefore only the percentage difference is reported for this artefact rather than an E_n score.

Descriptor	FN 111	FN 101
R_a	$66.3 \pm 4.7\,\mathrm{nm}$	$8.525 \pm 0.115\,\mu\mathrm{m}$
R_q	$77.6 \pm 5.0\,\mathrm{nm}$	$10.376 \pm 0.090\,\mu\mathrm{m}$
R_z	$278 \pm 22\,\mathrm{nm}$	$42.07 \pm 0.48\,\mu\mathrm{m}$
$RONt$	$278 \pm 22\,\mathrm{nm}$	$42.07 \pm 0.48\,\mu\mathrm{m}$
$RONq$	$77.6 \pm 5.0\,\mathrm{nm}$	$10.376 \pm 0.090\,\mu\mathrm{m}$
L^1 mean	$66.3 \pm 4.7\,\mathrm{nm}$	$8.525 \pm 0.115\,\mu\mathrm{m}$
L^2 RMS	$77.6 \pm 5.0\,\mathrm{nm}$	$10.376 \pm 0.090\,\mu\mathrm{m}$
L^∞	$150 \pm 12\,\mathrm{nm}$	$21.97 \pm 0.42\,\mu\mathrm{m}$
BV total	$1.137 \pm 0.062\,\mu\mathrm{m}$	$118.4 \pm 6.2\,\mu\mathrm{m}$
BV density	$0.263 \pm 0.014\,\mathrm{nm\,sample}^{-1}$	$6.58 \pm 0.35\,\mathrm{nm\,sample}^{-1}$
$W^{1,2}$ grad.	$0.310 \pm 0.015\,\mathrm{nm\,sample}^{-1}$	$8.76 \pm 0.43\,\mathrm{nm\,sample}^{-1}$
	ISO shape parameters	
R_{sk} (skewness)	-0.09 ± 0.04	0.00 ± 0.09
R_{ku} (kurtosis)	1.98 ± 0.21	2.51 ± 0.08
	Derived norm ratios	
L^4/L^1 peakiness	1.39 ± 0.08	1.53 ± 0.03
	External validation	
E_n score	0.61	< 1 (0.3% diff.; cf. Fig. S12)

FN 111, BV total variation carries 1.2% relative expanded uncertainty versus 5.5% for R_a and 8.5% for R_z, suggesting that gradient-based descriptors may exhibit metrological stability comparable to or better than peak-referenced roughness parameters in the tested datasets. Comparative bar charts and uncertainty analyses are provided in the Supplementary Materials (Figs. S7–S8). Ceramic and steel profile reconstruction plots are shown in Supplementary Figs. S9–S10.

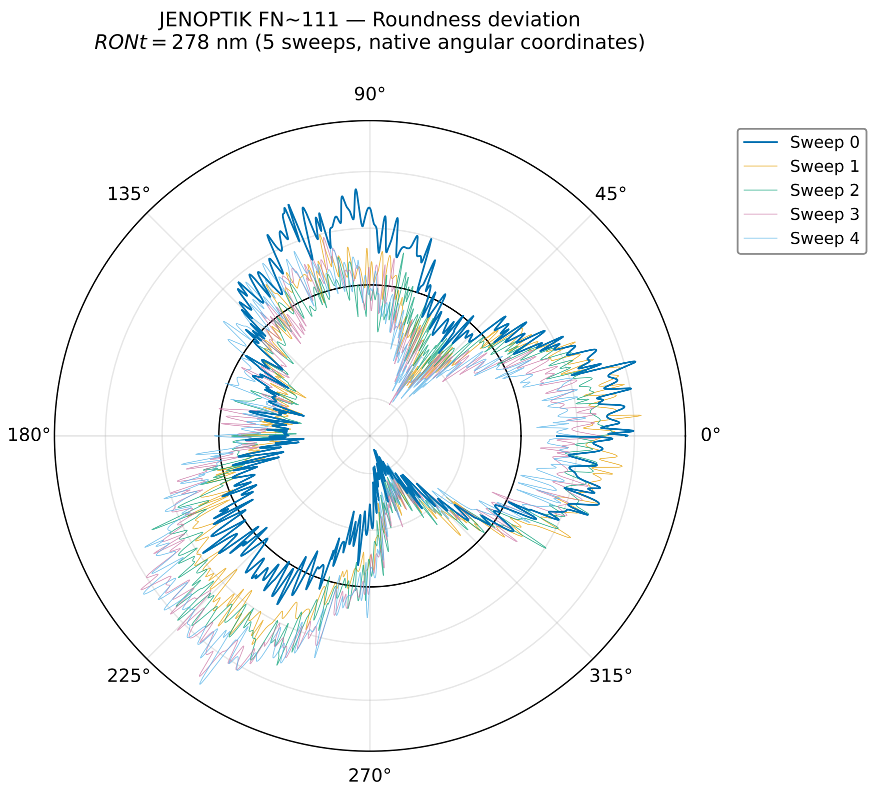

Figure 3: Ceramic roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native positions. $RONt = 278 \pm 22$ nm.

The steel roundness polar representation (FN 101, Flick flat) is provided in the Supplementary Materials (Fig. S11).

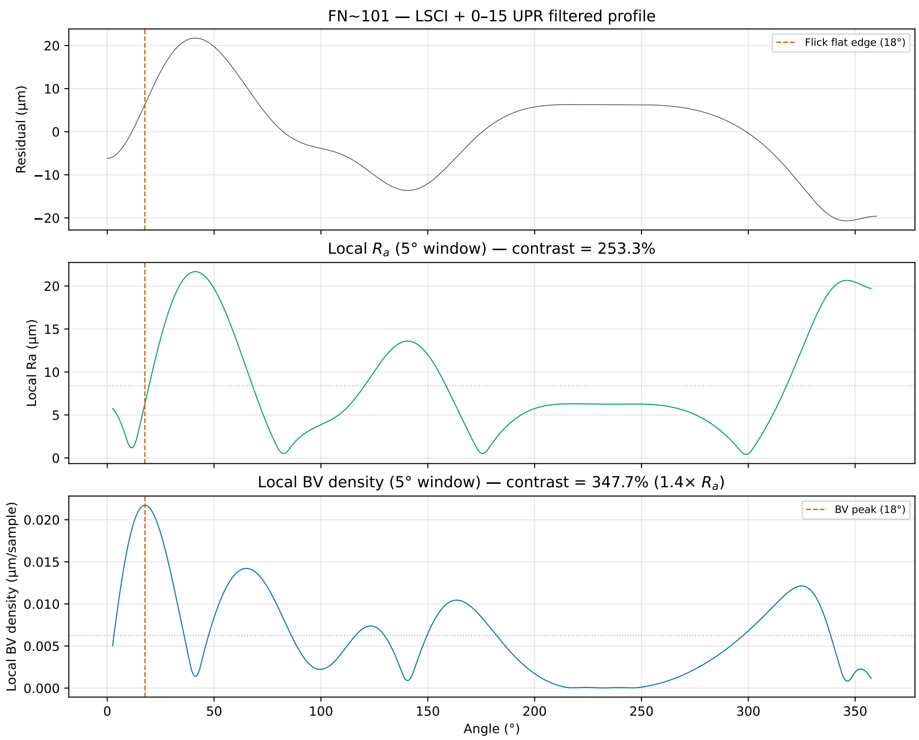

Figure 4: Flick flat detection on the FN 101 magnification standard. BV density (bottom) shows a clear spike at the Flick flat angular position, whereas R_a computed in a sliding window (top) shows no detectable response at the reconstruction resolution. The sliding-window width was matched to the certified angular extent of the Flick flat as specified in the DAkkS-DKD certificate (Art. 532 528); identical window parameters were used for both descriptors.

The Flick flat produces a clear response in BV density while showing no detectable response in a sliding-window R_a analysis at the reconstruction resolution (Fig. 4), illustrating the potential complementary sensitivity of gradient-based descriptors to localised geometric features.

5. Discussion

5.1. Confirmed Results

The analysis rests on a key metrological distinction: an interferogram is an intensity measurement, not a phase map and not a height map. The half-wavelength relation $z = \nu\lambda/2$ becomes valid only after each image has been decoded to a wrapped excess fraction and the ordered excess-fraction sequence has been unwrapped to the continuous fringe-order coordinate ν. This is why the reported comparison uses only reconstructed nanometre profiles, not raw greyscale images and not areal height maps.

The most sensitive part of the workflow is the EFM localisation of bright-ring maxima, with an excess-fraction uncertainty of approximately ± 0.05 fringe orders (tightened to ± 0.03 by sinusoidal refinement in 72% of FN 111 frames). A single ϵ_i error of 0.05 fringe corresponds to 13 nm, sub-dominant to the 85 nm residual-profile RMS. The chronological unwrapping criterion rejects phase jumps exceeding 0.35 fringe. The median number of detected rings is three for both artefacts within the 240 px central aperture. The five 400° ceramic sweeps require a geometric correction: successive useful 360° intervals are shifted by 40° because the physical object angle advances continuously; without this correction, the LSCI fit appears inconsistent even when phase extraction within each sweep is repeatable.

The linear drift of 0.0246 fringe frame^{-1} in the steel unwrapped phase sequence (versus 0.00021 for the FN 111) is attributed to cumulative bias in the global unwrapping algorithm rather than thermal effects or axial runout [45]. The lower fringe contrast of the FN 101 interferograms — likely due to surface finish — reduces per-frame EFM quality; the pipeline applies automatic linear detrending where needed.

5.2. Interpretation

5.2.1. Eccentricity removal and wobble verification

Roundness descriptors are computed after LSCI reference-circle fitting (Sec. 3.3), which removes the first harmonic. A shared model across all five sweeps yields

$R^2 = 0.99991$ with per-sweep amplitude variation of 63 nm (0.18%); the manufacturer-specified wobble (± 12 μrad) bounds the observed variation an order of magnitude below $RONt = 278$ nm. The polar representation (Fig. 3) shows five overlaid sweeps repeatable to ± 50 nm, illustrating that the residual is dominated by low-order form deviation rather than random roughness.

5.3. Comparison with ISO Parameters

A central question of this work is whether gradient-based functional descriptors provide additional descriptive sensitivity alongside classical ISO parameters when applied to the same reconstructed profiles. To answer this rigorously, every descriptor was computed on the same reconstructed nanometre residual profiles.

Mathematical equivalences. The L^1 norm is defined as $\frac{1}{N}\sum |p_i - \bar{p}|$, which is identical to the definition of R_a. The L^2 norm is $\sqrt{\frac{1}{N}\sum(p_i - \bar{p})^2}$, identical to R_q. Computing both on the same centred residual yields *exact numerical equality*: $R_a - L^1 = 0$ and $R_q - L^2 = 0$ to machine precision for both artefacts (Table 2). These equalities are not empirical findings; they are definitional consequences. Labelling the arithmetic mean of absolute deviations as L^1 rather than R_a provides no new metrological information and should not be presented as a methodological advance.

It is emphasised that neither the L^p norms nor the Sobolev and BV seminorms are presented as novel mathematical constructs; they are standard tools whose definitions are restated for completeness. Their value in the present context is threefold. First, the functional-analytic taxonomy places ISO parameters at specific positions (L^1, L^2, L^∞) within a larger space of possible descriptors, revealing that descriptors outside the ISO set — BV density, $W^{1,2}$ seminorm — respond differently to the same perturbations. Second, the synthetic-defect injection methodology provides a controlled, like-for-like comparison framework that can be applied to any descriptor computed from the same residual profile; the framework itself, rather than any individual norm, constitutes the methodological contribution. Third, the Flick flat on the certified FN 101 standard provides a concrete existence proof: a real, DAkkS-DKD-certified artefact whose defining geometric feature is detected unambiguously by BV density while remaining below the detection threshold of sliding-window R_a (Fig. 4).

Extrema: R_z versus L^∞. $R_z/L^\infty \approx 2$ for both artefacts, consistent with symmetric positive and negative excursions. L^∞ captures the single largest one-sided excursion; the ratio serves as a simple symmetry indicator.

Variation-based descriptors. A controlled sensitivity analysis (Supplementary Table S2) examined five synthetic defect types injected into the FN 111 residual profile. Amplitude-averaging descriptors (R_a, R_q, R_z) showed no detectable change at the reconstruction resolution — a single 100 nm spike changed R_a by less than 0.05%, below the per-frame phase-extraction precision — while BV density increased by +18% and the $W^{1,2}$ gradient seminorm by +622% for the same spike (Fig. 5). ISO shape parameters responded to asymmetry (R_{sk} changed by -1104% for a step) but not to symmetric localised features. R_{sk} and BV density appear complementary in the tested scenarios: R_{sk} responds to asymmetry, BV density responds to localised gradients.

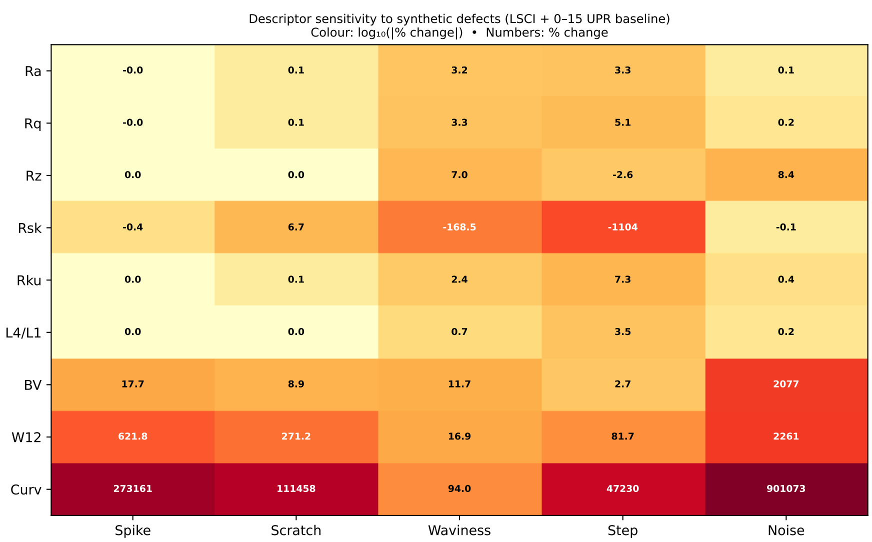

Figure 5: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Percentage change from the unperturbed baseline; bold labels indicate changes exceeding 10%.

Supporting analyses — descriptor correlation matrices, power spectral density comparison, Cohen's d effect sizes, critical descriptor assessment, and L^p norm convergence — are provided in the Supplementary Materials (Figs. S13–S17). Given the number of descriptors compared ($\sim$15 across two artefacts), the reported differences should be interpreted as exploratory rather than confirmatory; no formal multiple-comparison correction has been applied.

Practical considerations. A practical suggestion for interferometric profile and roundness metrology is to retain R_a, R_q, R_z, R_{sk}, $RONt$ and $RONq$ as the primary engineering layer compliant with ISO 4287, and to consider supplementing them with BV density (in angular units, $\mathrm{nm}\,\mathrm{deg}^{-1}$) as a gradient-sensitive functional-analysis complement. The $W^{1,2}$ gradient seminorm is correlated with BV density in the present datasets (within the FN 111: Pearson $r = 0.95$, $n = 5$ sweeps; within the FN 101: $r = 0.93$, $n = 5$). The pooled $r = 0.94$ (95% CI $[0.72, 0.99]$, $n = 10$) is reported for descriptive purposes only — the sweeps are not independent observations and this pooled correlation should be interpreted with caution. $W^{1,2}$ may add limited independent information beyond BV density alone given the small sample. The L^4/L^1 peakiness ratio provides a supplementary distribution-shape axis with a direct physical interpretation as the amplification of dominant profile excursions relative to R_a. These additional descriptors are computed from the same residual profile without additional hardware or changes to the EFM pipeline. The cross-instrument validation suggests that gradient-based descriptors are applicable to any 1D profile, whether from interferometry or tactile roundness measurement.

Planned versus exploratory comparisons. The synthetic-defect responses (Supplementary Table S2, Fig. 5) and the Flick flat detection (Fig. 4) were planned comparisons motivated by the study's objective of assessing gradient-based descriptor sensitivity. The correlation matrices, power spectral density comparison, Cohen's d effect sizes, L^p norm convergence analysis, and the cross-descriptor stability comparison (Supplementary Figs. S13–S17) are exploratory and are intended to characterise the descriptor space rather than to test specific hypotheses. This distinction should be borne in mind when interpreting the reported associations. The present study is a proof-of-concept demonstration; generalisability to other artefact types, materials, and measurement instruments remains to be established.

5.4. Limitations

Several limitations should be acknowledged. First, the uncertainty analysis is restricted to within-run Type A repeatability from five sweeps in a single continuous rotation. The small $n = 5$ limits statistical power (relative differences below $\sim 30\%$ cannot be resolved), and between-run reproducibility (independent setups, thermal cycles, re-centring) has not been assessed. A full GUM-compliant uncertainty budget, incorporating type-B evaluations of laser wavelength stability, refractive-index variations, Abbe error, camera nonlinearity, and form-removal uncertainty, is beyond the scope of this study; the reported expanded uncertainties should be interpreted as lower bounds.

Second, the validation is restricted to two artefacts from a single manufacturer (JENOPTIK), both ~ 30 mm diameter; generalisability to other sizes, materials, or surface finishes has not been tested. Third, the synthetic-defect analysis uses idealised mathematical defects (isolated spike, uniform scratch, pure sinusoid) injected into the FN 111 ceramic profile only; the response to real manufacturing defects with complex morphologies may differ, and the defect-to-background ratio for the FN 101 is approximately 500 times smaller, so the detection sensitivity reported here may not generalise to artefacts with substantially larger form deviations. Fourth, BV density and $W^{1,2}$ seminorm carry intrinsic sampling-interval dependence requiring conversion to angular units for cross-instrument comparison. Fifth, the Flick flat result (Supplementary Fig. S18) is demonstrated on a single artefact with a known macroscopic feature; performance for subtler, micrometre-scale defects has not been characterised.

5.5. Metrological Implications

The results carry several implications for profile and roundness metrology practice. The finding that BV density responds to localised gradients that amplitude averaging may not capture suggests — subject to further validation —

a potential role for gradient-based descriptors in applications where isolated defects are of concern; possible examples include quality control of bearing surfaces, optical components, or precision seals. Because these descriptors are computed from the same residual profile used for ISO 4287 parameters, their adoption does not require changes to existing measurement acquisition protocols.

The ISO 4287 framework remains the appropriate baseline for surface texture specification and conformity assessment. The gradient-based descriptors discussed here are proposed as optional supplementary indicators, not as replacements for standardised parameters. Their metrological utility would be strengthened by inter-laboratory comparisons, by evaluation on a wider range of artefact types, and by the development of agreed procedures for converting sampling-dependent units to instrument-independent angular gradient units. Until such steps are taken, BV density and related descriptors should be regarded as research-grade tools whose diagnostic value has been observed under controlled laboratory conditions on two specific artefacts.

5.6. Future Work

The present study suggests several directions for further investigation. (i) A between-run reproducibility study with independent setups and re-centring would establish whether the gradient-based descriptors retain their relative stability under realistic laboratory conditions. (ii) Extending the validation to artefacts with different diameters, materials (e.g., glass, carbide), and surface finishes would test the generalisability of the findings. (iii) A systematic characterisation of BV density response to real manufacturing defects (scratches, pits, inclusions) of controlled dimensions would bridge the gap between the present synthetic-defect analysis and industrial inspection scenarios. (iv) An inter-laboratory comparison using a common artefact set and a standardised processing pipeline would provide the reproducibility data needed to assess whether BV density could eventually be considered for standardisation. (v) The development of agreed angular-unit conventions for sampling-dependent gradient descriptors would facilitate cross-instrument comparison and potential inclusion in reference software for roundness evaluation.

Data Availability

The analysis code, processed results, and all figures are available at `https://github.com/dawidkucharski/banach-profile-metrology` (release v1.1.0). The raw interferometric PNG archives (∼32 GB) are available from the corresponding author upon reasonable request.

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